Urban Mobility Database for Traffic Pattern Analysis

Complete Project Explanation for Advanced SQL PBL-I

Project Title: Design and Implementation of an Urban Mobility Database for Spatio-Temporal Traffic Analysis Using Advanced SQL

Course: 21CSE742P, Advanced SQL and Modern Database Features

1. What are we actually building?

We are building a database system for urban traffic data. A city generates information about roads, vehicles, traffic sensors, road segments, trips, traffic volume, speed, congestion and time. Our job is to store this information properly in a relational database and use Advanced SQL to understand traffic patterns.

2. What does Urban Mobility mean?

Urban means city. Mobility means movement. Urban mobility therefore means how people and vehicles move around a city. In this project, the main focus is vehicle movement and traffic conditions.

3. What does Spatio-Temporal mean?

Spatial means WHERE, such as OMR Road, Anna Salai, a road segment or a sensor location. Temporal means WHEN, such as 8:00 AM, Monday or a particular date. Spatio-temporal therefore means WHERE + WHEN. A query can ask how much traffic occurred on OMR at 8 AM.

4. Why do we need a database?

A real traffic dataset can contain millions of records. We need a structured database to store the data efficiently, maintain relationships between entities, handle large numbers of records, perform complex queries, analyse traffic patterns and generate reports.

5. Main database tables

ROAD stores road_id, road_name, road_type and area. ROAD_SEGMENT stores segment_id, road_id, start_location, end_location and length. SENSOR stores sensor_id, segment_id and sensor_location. TRAFFIC_OBSERVATION is the central traffic data table and stores observation_id, sensor_id, timestamp, vehicle_count, average_speed and congestion_level. VEHICLE stores vehicle_id and vehicle_type. TRIP stores trip_id, vehicle_id, start_time, end_time, source and destination.

6. How are the tables connected?

ROAD has many ROAD_SEGMENT records. A ROAD_SEGMENT can have many SENSOR records. A SENSOR can produce many TRAFFIC_OBSERVATION records. A VEHICLE can have many TRIP records. Conceptually: ROAD $\rightarrow$ ROAD_SEGMENT $\rightarrow$ SENSOR $\rightarrow$ TRAFFIC_OBSERVATION, and VEHICLE $\rightarrow$ TRIP.

7. Where does Advanced SQL come in?

We are not only creating tables. We will use SQL to answer complex questions from the database, such as finding the busiest roads, identifying peak traffic hours, comparing traffic between locations, finding roads above average traffic, ranking roads and measuring query performance.

8. Example: finding the busiest roads

We can join ROAD, ROAD_SEGMENT, SENSOR and TRAFFIC_OBSERVATION and calculate average traffic for each road. This demonstrates multiple JOINs, AVG, GROUP BY and ORDER BY.

9. Example: finding roads above average

A subquery can calculate the overall average traffic and another query can return roads whose traffic is greater than that average. This demonstrates aggregation and subqueries.

10. Example: ranking roads

A window function such as RANK() OVER (ORDER BY traffic DESC) can rank roads from the busiest to the least busy. This demonstrates analytical SQL.

11. What is a CTE?

A Common Table Expression, or CTE, creates a temporary named result that can be used by a following query. We can first calculate average traffic for every road and then use that result to find roads above the overall average.

12. What are indexes?

If a traffic table contains millions of records, searching without an appropriate index can be slow. An index helps the database locate relevant records more efficiently, similar to the index of a textbook. Possible indexed columns include timestamp, sensor_id, segment_id and road_id. We can compare query execution before and after indexing.

13. What is a View?

A View is a reusable SQL query. If we frequently need a daily traffic report, we can create a View containing the required query and then query the View instead of rewriting the complete SQL each time.

14. What is a Materialized View?

A Materialized View stores the result of a query so that frequently requested analytical results can be reused more efficiently, depending on the database system. It can be useful for daily traffic summaries and other repeated reports.

15. What does Spatio-Temporal Traffic Analysis mean?

A spatial question is: Which road has the highest traffic? A temporal question is: What time has the highest traffic? A spatio-temporal question combines them: Which road has the highest traffic between 8 AM and 10 AM? We can also compare traffic on the same road across different days or time periods.

16. What data will we use?

We can use a traffic dataset containing fields such as sensor_id, timestamp, latitude, longitude, road_id, vehicle_count, average_speed and congestion_level. The raw dataset will be cleaned and transformed into our relational schema.

17. What is data cleaning?

Raw data may contain missing, invalid or inconsistent values. For example, a negative speed is invalid and an extreme speed value may need investigation. We will handle missing values, standardize dates and times, clean road names and prepare valid identifiers before loading the data into the database.

18. Complete project workflow

Traffic / Mobility Dataset → Data Cleaning and Preparation → Relational Database Design → Tables + Constraints + Relationships → Data Loading → Indexes + Views + Materialized Views → Advanced SQL Queries → Spatio-Temporal Traffic Analysis → Reports and Traffic Insights.

19. What makes this an Advanced SQL project?

The project uses multi-table queries, subqueries, CTEs, aggregate functions, window functions, indexing, views, materialized views and time-based analysis over a relational urban mobility database. The database and SQL work are the core of the project, not machine learning.

20. What is the research gap?

Previous research covers moving-object databases, traffic sensor analysis, spatio-temporal queries, indexing and data warehousing. Our project aims to combine these database ideas into one practical, student-level urban mobility database and demonstrate them through Advanced SQL.

21. What is our contribution?

We are not claiming to invent a new database technology. The contribution is a practical implementation that combines urban mobility data, relational database design, spatio-temporal traffic records, Advanced SQL queries, indexing, views, materialized views and historical traffic analysis.

22. What will the final system show?

The system can generate reports such as most congested roads, average speed by road, traffic volume by location, peak traffic hours, traffic trends by day and historical comparisons. We can also show query-performance comparisons before and after indexing.

23. Technologies

A practical stack can be Oracle or PostgreSQL for the database, a CSV traffic dataset for the source data, and Python only where useful for data cleaning or loading. Since the course emphasizes Oracle SQL, Oracle is a strong choice if the required environment is available.

24. What do I need to learn?

First learn database basics: tables, primary keys, foreign keys, relationships, normalization and constraints. Then learn SQL: SELECT, WHERE, GROUP BY, HAVING, JOINs, subqueries, CTEs and window functions. Then learn indexes, views, materialized views and transactions. Finally understand the project domain: roads, road segments, sensors, vehicles, trips, observations, timestamps and congestion.

25. How do the research papers fit?

The literature survey provides the foundation for the project. Different papers study moving-object databases, traffic sensor queries, spatio-temporal data, indexing and traffic pattern analysis. We use these studies to understand what has already been done, identify limitations and justify our proposed practical database workflow.

26. Questions your teacher may ask

Why did you choose this topic? Because urban traffic generates large amounts of structured data and it provides a practical problem for demonstrating Advanced SQL. Is this an ML project? No. The core is database design and Advanced SQL. What is spatio-temporal data? Data containing both spatial information, such as where an observation occurred, and temporal information, such as when it occurred. What is the research gap? Existing studies address moving-object databases, traffic sensor analysis, indexing and spatio-temporal queries separately. We aim to combine these concepts into a practical urban mobility database demonstrated using Advanced SQL. What are your main tables? Road, Road Segment, Vehicle, Sensor, Traffic Observation and Trip. What is the main table? Traffic Observation, because it stores actual traffic measurements collected from sensors over time. What Advanced SQL will you use? Joins, subqueries, CTEs, aggregate functions, window functions, views, materialized views, indexes and time-based queries.

27. The project in one example

Suppose an OMR sensor records 900 vehicles, an average speed of 18 km/h and high congestion at 8 AM. At 2 PM, the same sensor records 400 vehicles, an average speed of 42 km/h and low congestion. The database stores both observations. SQL can calculate the average traffic on OMR, identify peak hours, find the most congested roads, compare traffic with the city average and measure query performance after indexing.

28. The one-minute explanation to memorize

We collect traffic data, clean it, store it in a properly designed relational database, connect roads, sensors and observations, and use Advanced SQL to find traffic patterns across different locations and times. The project is NOT about predicting traffic using machine learning. It is about building a good database for traffic data and using

Advanced SQL to extract useful information from it.